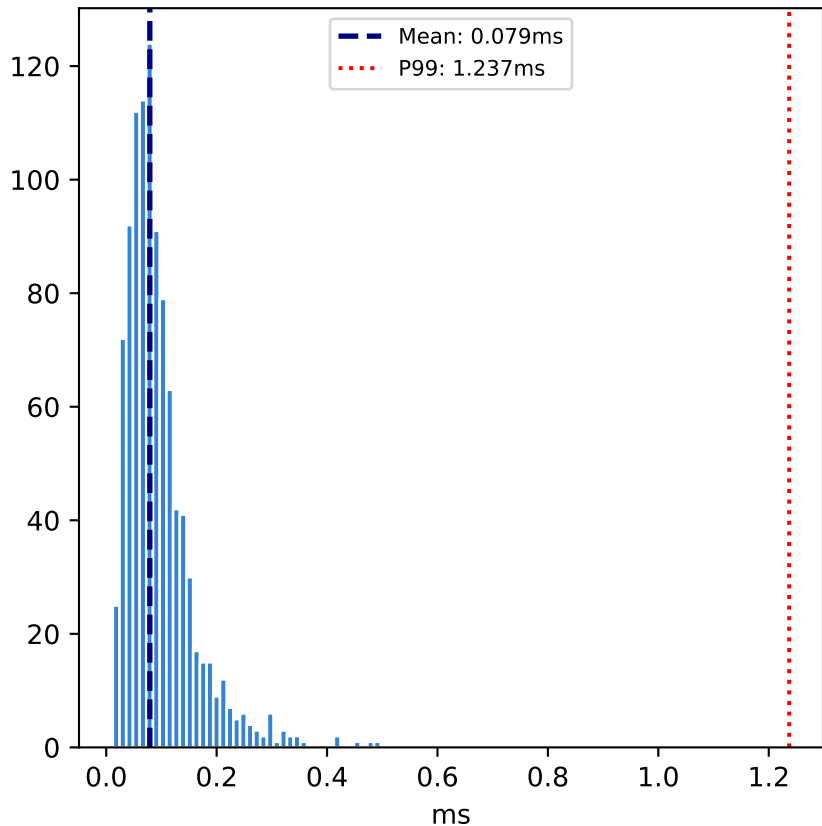
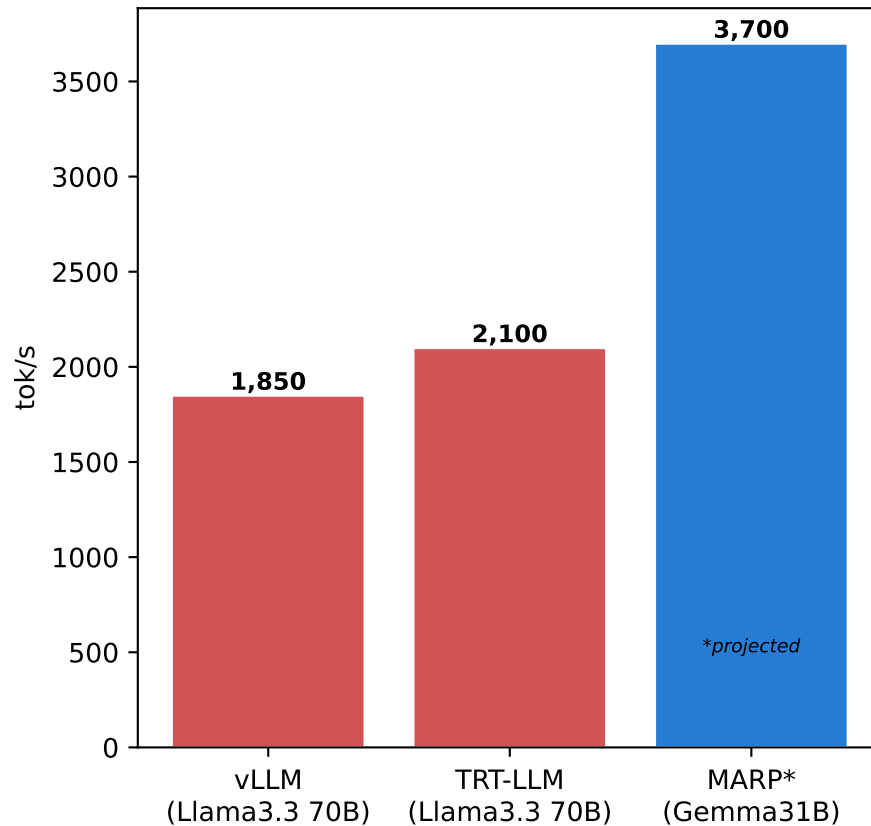


MARP Router — Local Benchmarks (100 queries, CPU-only)

Router Latency Distribution

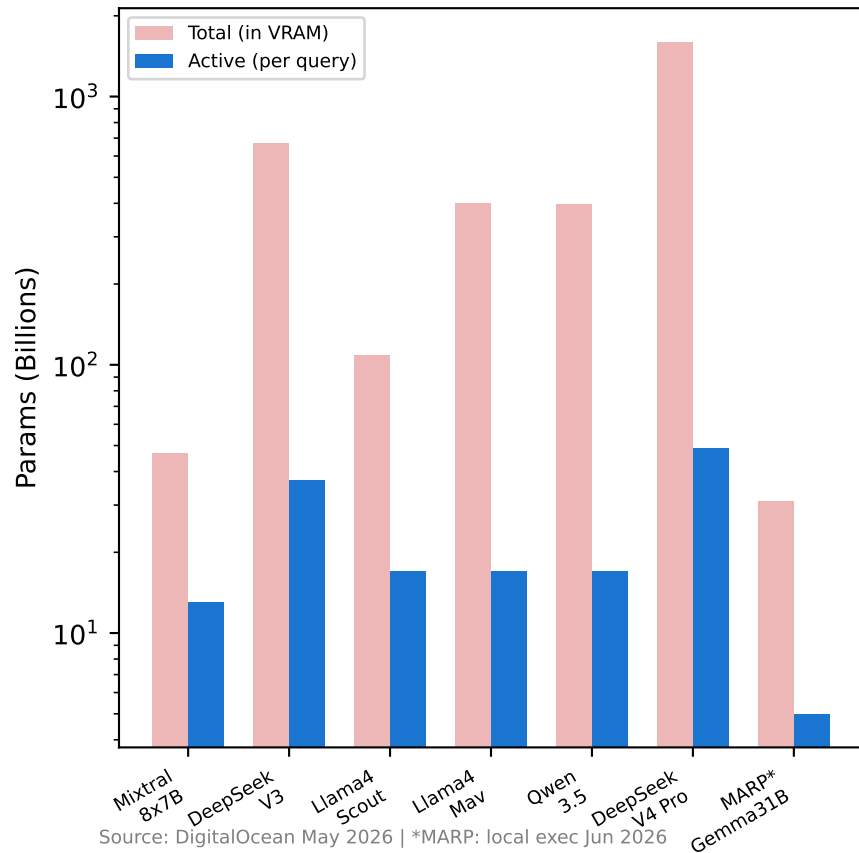


Throughput @50req (tok/s)

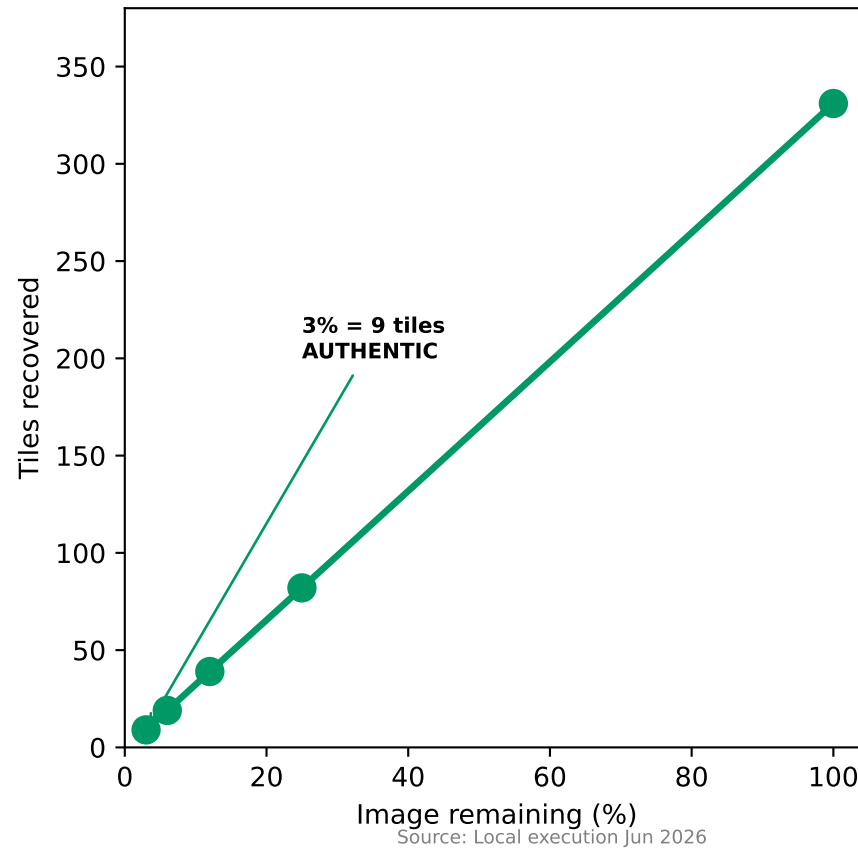


MoE Active/Total Params + H-Bit Crop Robustness

Active vs Total Parameters



H-Bit Spectrum: Crop Robustness (512x512 PNG, 98.3% confidence at 3%)



LOCALLY EXECUTED BENCHMARKS

TEST	TRIALS	RESULT	UNIT
MARP Router	100	0.079	ms
MARP Router	100	0.062	ms
MARP Router	100	1.237	ms
MARP Router	100	12626.0	q/s
MARP Router	25	0.36	ratio
MARP Router	25	0.622	ratio
H-Bit Spectrum	1	0.983	ratio
H-Bit Spectrum	1	331.0	tiles
H-Bit Spectrum	1	707.6	ms
H-Bit Spectrum	1	0.983	ratio
H-Bit Spectrum	1	82.0	tiles
H-Bit Spectrum	1	0.983	ratio
H-Bit Spectrum	1	39.0	tiles
H-Bit Spectrum	1	0.983	ratio
H-Bit Spectrum	1	19.0	tiles
H-Bit Spectrum	1	0.983	ratio
H-Bit Spectrum	1	9.0	tiles

EXTERNAL BENCHMARKS (cited from published sources)

- [1] Spheron, March 2026
spheron.network/blog/vllm-vs-tensorrt-llm-vs-sglang-benchmarks
 H100 80GB, Llama 3.3 70B FP8
 vLLM v0.18: 1,850 tok/s @50req, TTFT 120ms, VRAM 70GB
 TensorRT-LLM v1.2: 2,100 tok/s, TTFT 105ms
 SGLang v0.5.9: 1,920 tok/s, TTFT 112ms
- [2] DigitalOcean, May 2026
digitalocean.com/community/tutorials/mixture-of-experts-inference-cost
 MoE active/total ratios:
 Mixtral 8x7B: 13B/47B (27.7%) | DeepSeek V3: 37B/671B (5.5%)
 Llama 4 Scout: 17B/109B (15.6%) | Llama 4 Maverick: 17B/400B (4.3%)
 Qwen 3.5: 17B/397B (4.3%) | DeepSeek V4 Pro: 49B/1.6T (3.1%)
 'You pay for memory in total params, compute savings on active'
- [3] Signal65, December 2025
signal65.com/research/ai/from-dense-to-mixture-of-experts
 DeepSeek-R1: GB200 NVL72 28x perf vs MI355X, 1/15th cost-per-token
 Dense baseline: Llama 3.3 70B at 30-110 tok/sec/user
- [4] NVIDIA Public Specs
 H100 80GB (3,350 GB/s) | H200 141GB (4,800 GB/s)
 B200 192GB (8,000 GB/s) | RTX 3090 24GB (936 GB/s)
 DGX Spark 128GB unified memory